

Solution Requirements

Team ID	SWUID2026383049
Project Name	EduGenie: Google Gemini Powered Learning Assistant
Team Size	4
Team Leader	Ezhumalai V
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Functional Requirements

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Question Answering	User enters a question and receives a concise Gemini generated answer (/qa)
FR-2	Concept Explanation	User enters a topic and receives a simple explanation (/explain)
FR-3	Text Summarisation	User pastes a passage and receives a short summary (/summarize)
FR-4	Quiz Generation	System generates 3 MCQs with 4 options and checks answers (/quiz)
FR-5	Learning Path	System returns a beginner-to-advanced roadmap with resources (/learn/recommendations)
FR-6	API Key Configuration	User can check or update the Gemini API key from the UI (/api/config)

Non-Functional Requirements

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	Simple, responsive web interface usable by students of all levels
NFR-2	Security	API key stored in environment variables and never exposed in code
NFR-3	Reliability	Automatic fallback between Gemini models and to local LaMini-Flan-T5
NFR-4	Performance	Responses returned within a few seconds for typical queries
NFR-5	Availability	Runs locally with Uvicorn and can be deployed to any cloud host
NFR-6	Scalability	Modular design allows new learning modules to be added easily